

NENG LI

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TECHNICAL SKILLS

Languages: Python, Go, Rust, Java, C++, C, SQL, JavaScript, HTML/CSS, Bash Scripting

Frameworks/Libraries: PyTorch, HuggingFace, scikit-learn, NumPy, Pandas, FastAPI, Flask, Docker, Qdrant, Git, SLURM

Tools/Platforms: VS Code, GitHub, Vercel, Claude Code, Codex, Cursor, Agent Skills, MCP

EXPERIENCE

Research Assistant

May 2026 – Present

Castorini (Prof. Jimmy Lin), University of Waterloo | *Python, PyTorch, HuggingFace*

Waterloo, ON

- Co-built **1ShotBench**, a benchmark to evaluate coding agents' open-ended implementations, contributing a **LLM-as-a-judge** pipeline using **Playwright** to gather browser evidence; paper accepted to **CIKM 2026**
- Maintained **Tevatron**, a retrieval library, adding backbone-agnostic reranker support and sharded **FAISS** search
- Designed the architecture of a novel visual document retriever based on DeepSeek-OCR
- Ran benchmarking across ViDoRe v1/v3 and ColPali training datasets to guide experiments and tradeoffs for the visual document retriever, to make it a strong single-vector retriever (nDCG@10 = 0.76)

PROJECTS

🔗 SEC Filings RAG Pipeline | *Python, Qdrant, fastembed, Docker, XBRL*

- Built a hybrid dense/sparse **RAG pipeline** (**Qdrant**, bge-large-en-v1.5, BM25, cross-encoder reranking) over 150 SEC filings, reaching **78.9% accuracy** on FinanceBench
- Raised eval accuracy from **52% to 68%** by shipping **LLM-based query decomposition** after root-causing wrong answers to a cross-page synthesis failure mode
- Designed a **data-leakage eval** with post-training-cutoff filings, showing recall collapsed while RAG held at 68%+

🔗 NBA Game Predictor | *Python, scikit-learn, PyTorch, FastAPI, React, Streamlit*

- Built a leakage-safe **ML pipeline** ingesting 20+ NBA seasons via a rate-limited API, using retry/backoff and resumable checkpointing to reliably process 500K+ records
- Benchmarked **logistic regression**, **random forest**, and **LSTM** models in **scikit-learn/PyTorch** against market betting odds to quantify real-world model edge
- Shipped a **FastAPI + React** betting dashboard and a **Streamlit** research app for expected-value analysis

🔗 Silverfish – Chess Engine | *Go, Python, PyTorch, Python, Makefile, CI, SLURM*

- Built a **self-play training pipeline** to generate and label game data and fine-tune the neural network in **PyTorch**
- Reached **2400+ CCRL-calibrated rating** by adding **concurrent multi-threaded search** with lock-stripped caching
- Maintained project with **automated CI pipeline** with GitHub Actions and 49 unit/integration tests

🔗 Autonomous RL Race Car | *Python, Gymnasium, Matplotlib, Stable-Baselines3*

- Built an AI agent to play a 2D race car game using **reinforcement learning** (PPO) in Stable-Baselines
- Designed and optimized the **reward function** with waypoints, stagnation penalties, and crash avoidance
- Fixed slow/circular driving and crashes via **reward and environment shaping** to improve agent performance

EDUCATION

University of Waterloo

Bachelor of Software Engineering

Waterloo, ON

Sep 2025 – Present

- **GPA:** 3.97/4

- **Awards:** National Math Scholarship, President's Scholarship of Distinction